

ASSIGNMENT-3

GEM5



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GEM5 Simulation:Report

List of Applications to be simulated:

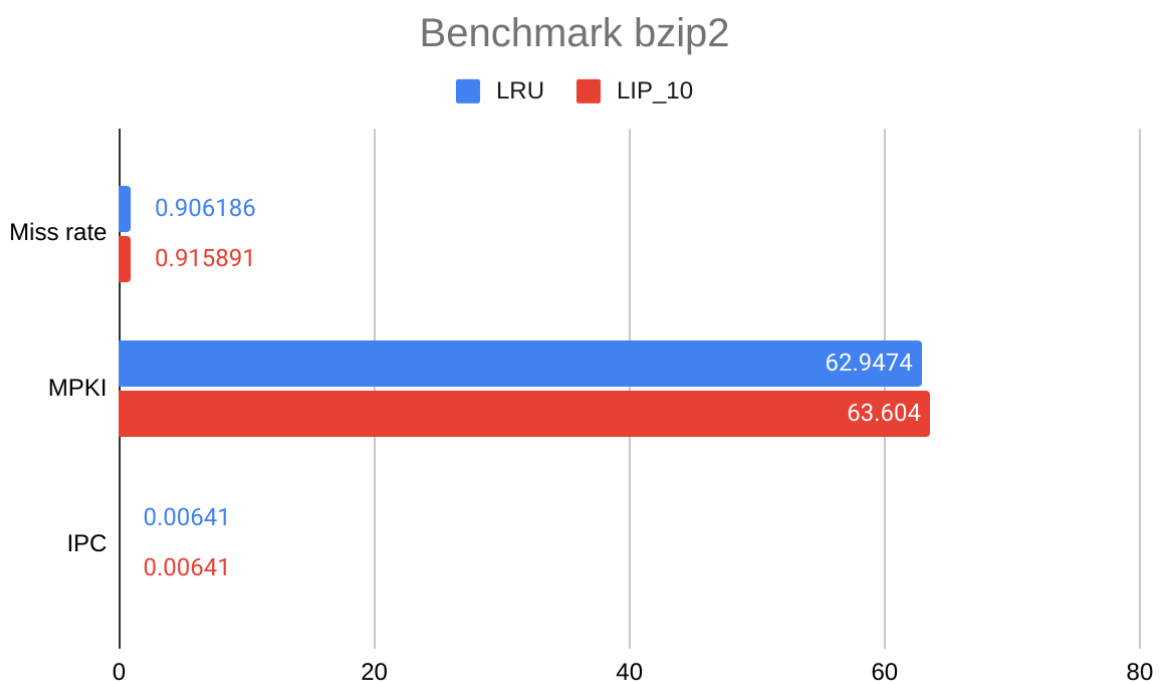
Bzip2 lbm namd gobmk(any three as mentioned)

LIP replacement policy:

LIP_10

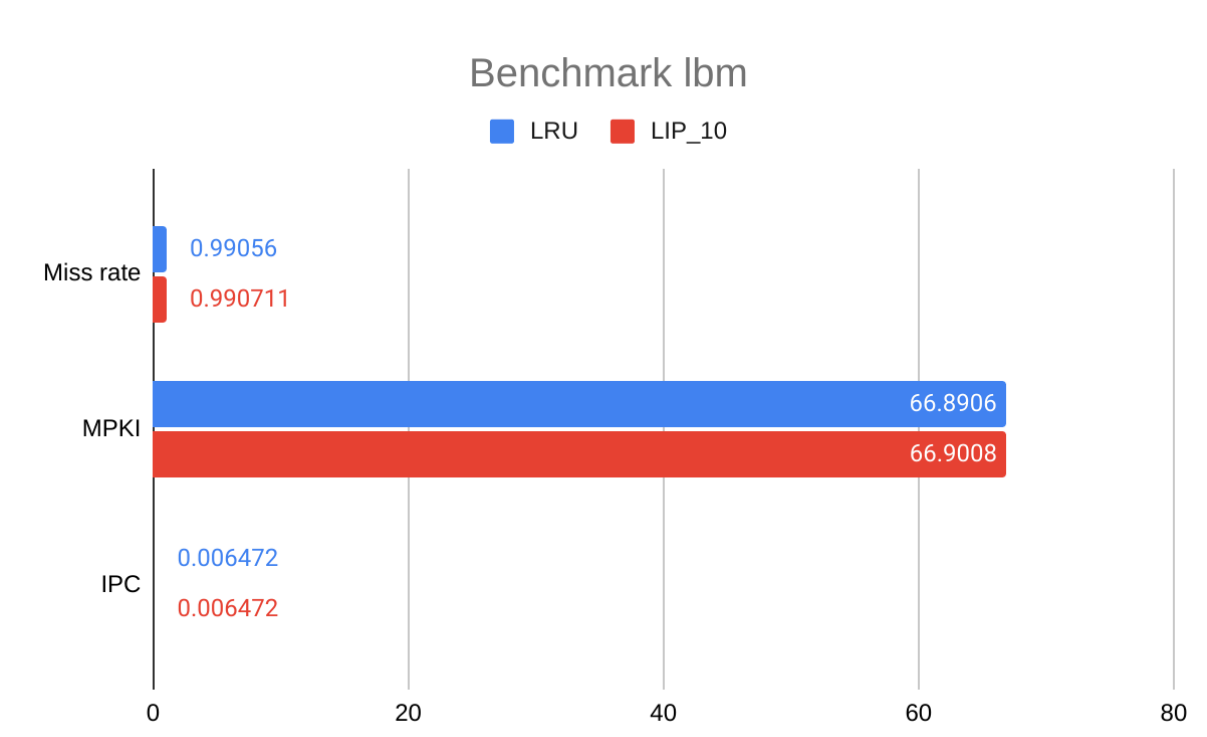
For benchmark bzip2:

bzip2	LRU	LIP_10
Miss rate	0.906186	0.915891
MPKI	62.9474	63.604
IPC	0.00641	0.00641



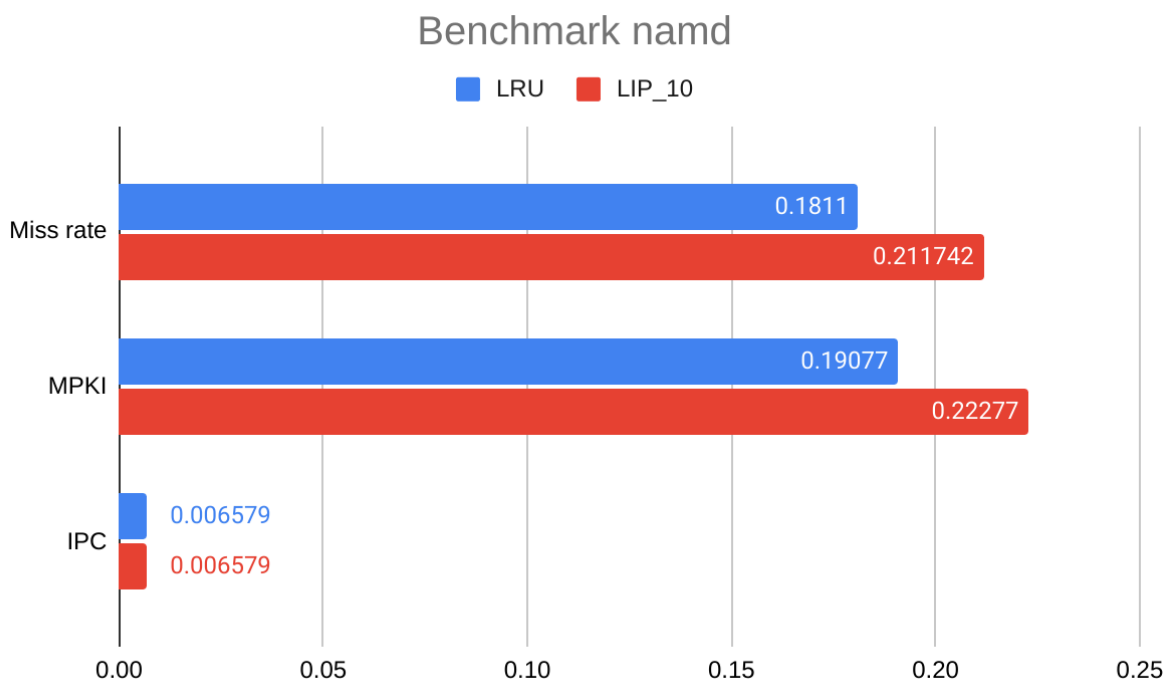
For benchmark lbm:

lbn	LRU	LIP_10
Miss rate	0.99056	0.990711
MPKI	66.8906	66.9008
IPC	0.006472	0.006472



For benchmark namd:

namd	LRU	LIP_10
Miss rate	0.1811	0.211742
MPKI	0.19077	0.22277
IPC	0.006579	0.006579



Conclusion:

- Using lbm there is nearly 16% reduction in miss rate for lru compared to lip_10 and there is less significance in other benchmarks, including better cache utilization of lru.
- In comparison of mpki, we can observe there is slightly better efficiency of lru compared to lip_10 on given benchcases.
- For every benchmark the ipc using lru and lip_10 are same.

Assumptions:

Some changes are made in Caches.py, and LIP_10 class .cc and .hh files are added and these files are uploaded.